

Workbook: Anatomy of the Cell

Week 1 workbook • BIO 004 Human Anatomy • Dr. Sharilyn Rennie • Solano Community College

A comprehensive workbook on the parts of a generalized human cell: the plasma membrane and its surface specializations, the nucleus, every major organelle, the cytoskeleton, and how a cell's organelle inventory predicts what it does for a living. Basic functions are included; molecular and biochemical mechanisms are not.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

The generalized cell • 2 questions

Q01 • DOK 1 • Recall

Name the three main regions of a generalized cell and what each contains.

Q02 • DOK 1 • Recall

Roughly how big is a typical human cell, and what controls the upper limit of cell size?

The plasma membrane • 7 questions

Q03 • DOK 1 • Recall

Describe the plasma membrane using the fluid mosaic model.

Q04 • DOK 1 • Recall

What does a phospholipid look like, and which end faces the cytoplasm and the extracellular fluid?

Q05 • DOK 1 • Recall

What is cholesterol doing in the plasma membrane?

Q06 • DOK 2 • Compare

Compare integral and peripheral membrane proteins by location and typical role.

Q07 • DOK 1 • Recall

What is the glycocalyx, what is it made of, and what does it do?

Q08 • DOK 1 • Recall

List the major functions of the plasma membrane.

Q09 • DOK 1 • Recall

Name three surface specializations of the plasma membrane and what each does.

The nucleus • 5 questions

Q10 • DOK 1 • Recall

What is the nucleus, what does it contain, and what does it do for the cell?

Q11 • DOK 1 • Recall

Describe the nuclear envelope.

Q12 • DOK 1 • Recall

What are nuclear pores and what passes through them?

Q13 • DOK 1 • Recall

What is chromatin, what is it made of, and how does it differ from a chromosome?

Q14 • DOK 1 • Recall

What is the nucleolus, and what is it doing?

Membranous organelles • 8 questions

Q15 · DOK 1 · Recall

Describe rough endoplasmic reticulum and what it does.

Q16 · DOK 1 · Recall

Describe smooth endoplasmic reticulum and what it does.

Q17 · DOK 1 · Recall

Describe the Golgi apparatus and what it does.

Q18 · DOK 1 · Recall

What are lysosomes, what do they contain, and what do they do?

Q19 · DOK 1 · Recall

What are peroxisomes, and what do they do?

Q20 · DOK 1 · Recall

Describe the structure of a mitochondrion and its basic function.

Q21 · DOK 2 · Apply

Mitochondria carry their own DNA and ribosomes. What does that tell you about their evolutionary origin, and why does it matter for inheritance?

Q22 · DOK 1 · Recall

What is a transport vesicle, and what does it do?

Non-membranous organelles and the cytoskeleton · 4 questions**Q23 · DOK 1 · Recall**

What are ribosomes, what are they made of, and where do they work?

Q24 · DOK 1 · Recall

What are centrioles, where are they found, and what do they do?

Q25 · DOK 1 · Recall

Name the three components of the cytoskeleton and what each one does.

Q26 • DOK 2 • Compare

Compare microfilaments, intermediate filaments, and microtubules by their protein, their diameter, and the kinds of jobs they handle.

The secretory pathway • 3 questions**Q27 • DOK 2 • Apply**

Trace the path of a secreted protein from gene to outside the cell, naming each structure it passes through.

Q28 • DOK 1 • Recall

What is endocytosis, and name two forms of it.

Q29 • DOK 1 • Recall

What is exocytosis, and what kinds of materials does the cell typically release this way?

Form follows function • 4 questions

Q30 · DOK 2 · Apply

A cell is packed with rough endoplasmic reticulum, ribosomes, and a prominent Golgi apparatus. Predict what kind of cell it is.

Q31 · DOK 2 · Apply

A cell is packed with mitochondria. What kind of work do you predict that cell does for a living?

Q32 · DOK 2 · Apply

A liver cell is packed with smooth endoplasmic reticulum. Predict why.

Q33 · DOK 3 · Reason

In Tay-Sachs disease, a single lysosomal enzyme is missing, and a specific lipid accumulates inside neurons. Using what you know about lysosomes, predict why the disease damages neurons.
